MoorSafe

INFO PACKET · SPECS, TESTS, INSTALLATION

The mooring anchor that won't chain wrap.



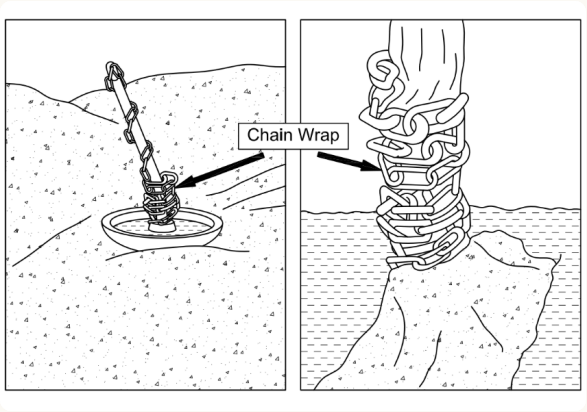
A conical, patent-pending redesign of the mushroom anchor — tested at Yale Cordage, proven in Casco Bay, made in Maine.

40,657 lb	Patent pending	Made in Maine
Tensile strength to failure	U.S. design patent filed	Designed, built & tested

01 · THE PROBLEM

Every mushroom anchor in a soft-bottom harbor has the same flaw.

The cast-iron mushroom anchor sinks into mud and gains suction over time — but its vertical stem is its weakness. As tides, currents, and wind swing a moored boat through 360°, the chain rotates around the stem. Once it catches, every cycle pulls it tighter. Effective scope shrinks, peak loads spike, mooring balls pull under at high tide, and chain or shackle failure becomes a question of when, not if.



Traditional mushroom — the stem becomes a winch.

THE MOORSAFE FIX

Three bars. Conical shape. Far less wrap risk.

Three symmetrically positioned bars rise from the bowl to a single point above the stem, forming a cone. The chain has nothing to catch on. The bowl still beds in, still gains suction, still holds — and tensile strength is higher than the original.

02 · SPECIFICATIONS & SIZING

Built like the original. Just smarter.

MODEL	RECOMMENDED FOR	CONDITIONS
400 lb	Boats ~30–40 ft	Heavier displacement or moderate exposure

MATERIAL
Cast-iron base, welded steel bars

FINISH
High-visibility safety orange

PATENT / ORIGIN
Patent pending · Maine, USA

Sizing depends on displacement, scope, and exposure. Pricing is by quote and excludes chain, hardware, and installation; shipping is quoted by destination. We'll confirm the right size for your mooring before you order.

03 · TEST RESULTS

Two tests. Both of them honest.

STRUCTURAL · YALE CORDAGE, SACO, ME

Tensile strength to failure

Horizontal tensile test on a 100-ft bed rated to 1.3 million lb. A 300 lb traditional mushroom and a 300 lb MoorSafe prototype loaded in tension until failure.

38,560 lb
Traditional · stem fracture

40,657 lb
MoorSafe · joint dislocation

> MoorSafe outperformed the traditional anchor by roughly 5%.

FIELD · CASCO BAY, ME

Breakout after 28-day embedment

Both anchors set on the same muddy seabed for 28 days, then pulled by a lobster boat on a 75-ft chain at 3:1 scope; tension read on a 4,000 lb digital indicator.

1,575 lb
Traditional · partially buried

1,295 lb
MoorSafe · fully buried

> Comparable breakout; MoorSafe was fully buried — holding improves as sediment consolidates.

04 · INSTALLATION

Installs exactly like the anchor it replaces.

1
Size it
Match weight to boat length, displacement, scope, and exposure — or ask us.

2
Rig it
Same chain, shackles, swivel, and pennant you use today. Attach at the top eye.

3
Set it
Lower with a mooring barge or service boat, same scope and practice as a mushroom.

4
Let it bed
Holding grows as the bowl embeds. Inspect on your normal schedule, as you would any mooring.

Any mooring service provider will be familiar with the process. We can recommend providers in your area.

05 · WHO WE ARE

Designed by people who know the harbor.

MoorSafe was founded by Scott Karkos and Captain Gregory Smith (1988–2025), a Maine Maritime Academy captain with two decades of mooring service work. We design and manufacture in Maine because the people who make our anchors are the same people whose boats depend on them.

NEXT STEPS

Put a MoorSafe on your mooring this season.

Request a quote: moorsafe.org/contact

Quote or questions: hello@moorsafe.com

Web: moorsafe.org

Harbormasters and mooring services: ask about fleet pricing and association presentations.